FOR EDITH · PAEDIATRICS & CHILD HEALTH JUNIOR CLERKSHIP

# Febrile Convulsions

Tutorial | Week 3 · Fits, TB, kidneys & nutrition | Paediatrics | Dr Nabbale

A febrile convulsion is a seizure with a fever of 38°C or more in a child aged about 6 months to 5–6 years, with no CNS infection, metabolic cause or previous afebrile seizure; it is a diagnosis of exclusion, so in Uganda every fitting febrile child needs glucose, a malaria test and a decision about meningitis before anyone says "just a febrile fit".

**Must know before the ward round**

1. Febrile convulsion: seizure + fever ≥38°C, age ~6 months to 5 years (UCG 2023 says 3 months–6 years), no CNS infection, no metabolic cause, no previous afebrile seizure. A diagnosis of exclusion.
2. Simple = generalised, <15 min, only once in 24 h (same illness), full recovery within 1 h. Complex = focal, ≥15 min, recurs within 24 h, or slow recovery/Todd's paresis. Febrile status = ≥30 min.
3. Treat any seizure still going at 5 minutes (ILAE 2015: t1 = 5 min). First line: diazepam 0.5 mg/kg rectally (max 10 mg) or buccal midazolam 0.3 mg/kg (max 10 mg); IV diazepam 0.2–0.3 mg/kg if a line is in.
4. Never more than 2 doses of benzodiazepine (count the dose given at home or at the HC). Then a second-line drug.
5. Second line: phenobarbital 15–20 mg/kg IV over 15–20 min (max 1 g) or IM (WHO 15 mg/kg), or phenytoin 15–20 mg/kg IV in saline at ≤1 mg/kg/min (UCG/WHO: over 60 min). Levetiracetam 40 mg/kg IV (max 3 g) where available (NICE NG217, 2022).
6. Neonates (<2 weeks to 1 month): phenobarbital 20 mg/kg IV/IM first, not diazepam; extra 10 mg/kg after 30 min, max 40 mg/kg.
7. Check glucose in every fitting child. Hypoglycaemia (UCG <3.0 mmol/L; WHO <2.5, <3.0 in SAM): 10% dextrose 5 mL/kg IV, recheck in 30 min.
8. In Uganda the commonest cause of fever with a fit is malaria. Malaria + 2 or more convulsions in 24 h = severe malaria (UCG 2023): IV artesunate 3 mg/kg if <20 kg (2.4 mg/kg if ≥20 kg) at 0, 12, 24 h then daily.
9. LP if meningism, bulging fontanelle, complex seizure, not back to normal within 1 h, age <12 months (lower threshold <2 years per UCG), incomplete Hib/PCV, or antibiotics already given. Not if signs of raised ICP or shock: give ceftriaxone 100 mg/kg/day and do the LP later.
10. Recurrence about 30% overall (50% if first fit before 1 year). Risk factors: age <18 months, family history, low peak temperature (<40°C), short fever before the fit (<1 h). All four >70%; none <20%.
11. Epilepsy risk after simple febrile seizures ≈1–2% (background ≈1%); higher after complex seizures, with a family history of epilepsy or developmental delay.
12. Antipyretics and tepid sponging do NOT prevent febrile convulsions (NICE NG143); no routine EEG, CT or long-term anticonvulsant after a simple febrile seizure.


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# In a nutshell

A **febrile convulsion** (febrile seizure) is a seizure that happens with a fever in a young child whose brain is otherwise normal and who has **no infection of the brain or meninges**. It is the **commonest seizure of childhood**: 2–5% of children have one, usually between **12 and 18 months** of age. A **simple** febrile convulsion is frightening to watch but **harmless**: it does not damage the brain, cause learning problems or (much) increase the risk of epilepsy.

The challenge in Uganda is that a child with fever and a fit is **just as likely to have cerebral malaria, bacterial meningitis or hypoglycaemia** as a benign febrile convulsion. So the diagnosis is made by **exclusion**: stop the fit, check the glucose, test for malaria, decide whether the child needs a lumbar puncture, and only then reassure the family.

> **CLINICAL PEARL**
> The one thing never to forget: **"febrile convulsion" is the last diagnosis you make, not the first.** Every fitting febrile child at Mengo gets ABC, a glucose check, a malaria test and a meningitis decision. A child who is **not back to normal within an hour** does not have a simple febrile convulsion.

# Definitions & classification

| Term | Definition |
|---|---|
| **Febrile convulsion** | Seizure with fever **≥38°C** in a child aged about **6 months to 5 years** (AAP; RCH Melbourne uses 6 months to 6 years; **UCG 2023 says 3 months to 6 years**), **without** CNS infection, metabolic disturbance (glucose, sodium, calcium), or a history of afebrile seizures. ILAE: after 1 month of age, with no previous neonatal or unprovoked seizure |
| **Simple febrile convulsion** (about 70–75%) | **Generalised** tonic–clonic, lasts **<15 minutes**, does **not recur within 24 h** (or the same illness), and the child is **fully recovered within 1 hour** |
| **Complex febrile convulsion** (about 25–30%) | **Any one** of: **focal** onset or focal features (including Todd's paresis afterwards); **≥15 minutes**; **recurrence within 24 h** or the same illness; prolonged altered consciousness (>1 h) |
| **Febrile status epilepticus** | A febrile seizure, or a series without recovery in between, lasting **≥30 minutes** (about 5% of febrile seizures) |
| **Status epilepticus (ILAE 2015 operational)** | Convulsive seizure lasting **>5 minutes** (time t1, when treatment must start) or repeated seizures without return to baseline; damage becomes likely after **30 minutes** (t2). **UCG 2023** still defines it as >30 minutes, repeated fits without recovery, or **not responding to 2 doses of diazepam**; treat at 5 minutes anyway |
| **Febrile seizure plus / GEFS+** | Febrile seizures continuing after 6 years, or with afebrile seizures, in a family with a genetic epilepsy (e.g. *SCN1A*); **Dravet syndrome** is the severe end (prolonged, often focal, febrile seizures from about 6 months) |

UGANDA CONTEXT

**UCG 2023** describes febrile convulsions as generalised, "usually <5 minutes, always <15 minutes", from **3 months to 6 years**, and lists **cerebral malaria, meningitis and epilepsy** as conditions to exclude first. It advises considering **LP or empirical meningitis treatment** when intracranial infection cannot be excluded clinically, **especially under 2 years**. International guidelines (AAP, NICE, RCH) start at 6 months. In a Ugandan ward, treat a child under 6 months with fever and a fit as **sepsis/meningitis until proven otherwise**.

# Epidemiology & Uganda context

- **Frequency:** 2–5% of children in Europe and North America by age 5 (higher in Japan, 6–9%); a sibling's risk rises to about 1 in 5, and higher still if a parent was affected.
- **Age:** peak **12–18 months**; uncommon before 6 months and after 5 years. Slight male predominance.
- **Triggers worldwide:** viral infections (HHV-6/roseola, influenza, adenovirus), otitis media, tonsillitis, gastroenteritis, and fever after some vaccines (MMR at 8–14 days, DTP within 48 h). Vaccine-related febrile seizures are uncommon and **not a reason to stop immunising**.
- **Uganda:** the leading cause of fever with convulsions in children admitted to Mulago and Mengo is **malaria**, followed by **pneumonia, URTI/otitis, UTI, gastroenteritis, bacterial meningitis and sepsis**. Hospital studies from East Africa show that a substantial share of "febrile seizures" are in fact **acute symptomatic seizures** from malaria, meningitis or hypoglycaemia, and that **complex seizures are commoner** than in Western series. Children often arrive after a **long journey**, after **several fits**, and sometimes after traditional treatments (herbs by mouth, incisions, smoke), which raises the risk of aspiration and burns.

UGANDA CONTEXT

Typical path: the child fits at home, is taken to a drug shop or HC II/III (perhaps given rectal diazepam or quinine IM), then to a HC IV or Mengo/Mulago. **Ask what was given and when**: a dose of diazepam given before arrival counts toward the maximum of two, and prior antibiotics change how you read the CSF.

# Pathophysiology

**Why do young brains fit with fever?**

1. **Immature brain:** between 6 months and 5 years the balance between excitatory (glutamate) and inhibitory (GABA) signalling favours excitation. The seizure threshold is low and is lowered further by temperature.
2. **Temperature and cytokines:** fever itself (and the **rate of rise**) increases neuronal excitability. Interleukin-1β and other cytokines released during infection are pro-convulsant. That is why fits often happen **early in the illness**, sometimes before the parents notice the fever.
3. **Genetics:** strong family clustering; several loci (FEB genes) and sodium-channel genes (*SCN1A*, *SCN1B*) in GEFS+ families.
4. **Prolonged seizures harm:** a seizure lasting more than about 30 minutes causes neuronal injury (excitotoxicity, hypoxia, hypoglycaemia, hyperthermia). **Febrile status** is linked to **hippocampal injury and later mesial temporal sclerosis**, which is why we treat at 5 minutes.

**Why do benzodiazepines become less effective with time?** As a seizure continues, GABA-A receptors are internalised from the synapse and NMDA receptors move to the surface. Benzodiazepines work on GABA-A receptors, so the **longer you wait, the less they work**. Give the first dose promptly and correctly, and move to a second-line drug after **two** doses.

**Why malaria fits differ:** *P. falciparum* sequesters in cerebral capillaries, causing microvascular obstruction, brain swelling, hypoglycaemia and metabolic acidosis. Seizures in cerebral malaria are frequent, often **subtle** (eye deviation, twitching of a finger or mouth, irregular breathing) and recur. They are acute symptomatic seizures, not febrile convulsions.

# Clinical acumen

## Focused history

Ask an eyewitness (the mother, not the person who carried the child last):

- **Describe the event**: stiffening and jerking of all four limbs (generalised) vs one side, eyes turned to one side, or one limb (focal = complex). Did it start on one side?
- **Duration**: parents overestimate; anchor with "how long to get from home to the HC?" ≥15 min = complex.
- **Number of fits** in this illness and in the last 24 hours (≥2 = complex; with malaria, **≥2 in 24 h = severe malaria** by UCG).
- **Recovery**: back to normal within an hour? Weakness of a limb afterwards (Todd's paresis)?
- **Fever**: how long before the fit? (short duration predicts recurrence). Highest temperature measured.
- **Source of fever**: cough, fast breathing, ear discharge, diarrhoea and vomiting, rash (measles), dysuria/foul urine, joint or bone pain.
- **Red-flag symptoms for meningitis/encephalitis**: irritability, high-pitched cry, refusing feeds, vomiting everything, headache, neck pain, photophobia, drowsiness, a rash, illness for several days before the fit.
- **Treatment already given**: antimalarials, **antibiotics** (partially treated meningitis), **diazepam** (dose and time), traditional medicine, anything put in the mouth.
- **Past history**: previous febrile or afebrile fits, neonatal problems (asphyxia, jaundice, admission), head injury, **sickle cell disease** (stroke), known epilepsy or cerebral palsy.
- **Development**: milestones normal for age? Regression? (abnormal development makes the diagnosis less likely and raises later epilepsy risk).
- **Family history**: febrile seizures, epilepsy.
- **Immunisation**: UNEPI card; specifically **Hib (DPT-HepB-Hib) and PCV** at 6, 10, 14 weeks. Incomplete → higher risk of bacterial meningitis → lower threshold for LP.
- **HIV status** of mother and child, and **feeding** (breastfeeding, weaning diet), **nutrition** (weight loss, oedema). HIV-exposed or malnourished children get meningitis, cryptococcus and TB more often.
- **Poisoning**: access to paraffin, organophosphates, traditional herbs, adult medicines (e.g. isoniazid, antimalarials, tricyclics).

## Examination

- **ABCD first** (ETAT): airway, breathing, circulation, **coma** and **convulsion**, **dehydration**. Glucose ("Don't Ever Forget Glucose").
- **Vital signs** with age norms: temperature, RR, HR, CRT, $SpO_2$, BP. Fast breathing: ≥50/min at 2–12 months, ≥40/min at 1–5 years.
- **Level of consciousness**: AVPU, and the **Blantyre coma scale** (pre-verbal children) or modified GCS. A child who remains **P or U** an hour after the fit is not a simple febrile convulsion.
- **Meningism**: neck stiffness, Kernig and Brudzinski signs (unreliable under 18 months), **bulging fontanelle** (best sign in infants, check with the child upright and quiet), photophobia, irritability when handled ("paradoxical irritability").
- **Raised ICP** (contraindicates LP): unequal or poorly reactive pupils, abnormal posturing (decorticate/decerebrate), focal limb weakness, irregular breathing, bradycardia with hypertension,

papilloedema (late), deteriorating coma score.

- **Skin**: **non-blanching petechiae/purpura** (meningococcus), measles rash, jaundice (malaria/haemolysis), pallor (malaria, sickle cell), neurocutaneous signs (café-au-lait macules, ash-leaf macules).
- **Focus of infection**: ears (otitis media), throat, chest, abdomen (spleen), joints, skin, urine.
- **Neurology** once awake: tone, power and symmetry of limb movements, reflexes, gait if walking, **head circumference**.
- **Growth**: weight, length/height and **MUAC**; plot **Z-scores** (WAZ, HAZ, WHZ). SAM changes fluids, glucose threshold and antibiotics.

### Red flags

> **RED FLAG**
> Not a simple febrile convulsion (treat as meningitis, cerebral malaria or another serious cause until proven otherwise):
>
> - Age **<6 months** (and especially <1 month)
> - Seizure **≥15 minutes**, **focal**, or **more than one** in 24 h
> - **Not fully alert within 1 hour**; Blantyre score ≤4 after the post-ictal period
> - **Bulging fontanelle**, neck stiffness, petechial/purpuric rash
> - Signs of **raised ICP** or focal neurological deficit
> - Fever for **several days** before the fit, or already on antibiotics
> - **Respiratory distress, shock, severe pallor** (malaria, sepsis), hypoglycaemia
> - **Developmental regression**, abnormal neurology, or known neurological disease

### Thinking like a clinician

- **"Fit + fever + malaria-positive" does not mean "malaria explains everything".** Up to a third of Ugandan children carry parasites with another illness. A positive RDT does **not exclude meningitis**; if there are meningeal signs or the child stays drowsy, do the LP (or treat for both).
- **Post-ictal or comatose?** After a simple fit the child is sleepy but rousable and improves within the hour. A Blantyre score that **stays low** means cerebral malaria, meningitis/encephalitis, hypoglycaemia, ongoing **subtle seizures** or poisoning.
- **Subtle seizures:** in a drowsy child with malaria watch for eye deviation, nystagmus, lip-smacking, finger twitching or irregular breathing. These are seizures and need treatment.
- **Rigors vs fits:** a febrile child shivering with a rigor is conscious, responds and the shaking stops when you hold the limb. A seizure involves loss of awareness and rhythmic jerking that does not stop with restraint.
- **Febrile syncope / breath-holding:** brief, child pale and floppy, recovers in seconds; may have a few jerks (reflex anoxic seizure).
- **Under 6 months:** fever and a fit = sepsis/meningitis workup and IV antibiotics. Under 1 month: neonatal sepsis protocol, phenobarbital first.
- **The second fit:** a child with a first fit at home and a second on the ward has a complex febrile seizure by definition: think harder, and do the LP if there is no contraindication.
- **Two benzodiazepine doses and still fitting = status**: move to phenobarbital/phenytoin, don't give a third diazepam dose (respiratory depression with little added benefit).

## Clinical workup

The fitting child is managed and assessed at the same time. The steps below are what the team leader checks while the treatment runs.

## Step 0: First 5 minutes

1. **Note the time** the fit started (ask the parent) and call for help.
2. **Airway**: position on the side (recovery position), suction the mouth gently, nothing in the mouth, no restraint. Jaw thrust if needed.
3. **Breathing**: oxygen by face mask or nasal prongs ($SpO_2$ target ≥94% during the fit; ≥90% once stable). Bag-mask ventilation if apnoeic.
4. **Circulation**: CRT, pulse, BP if possible; IV or IO access. Draw blood for **glucose**, malaria RDT/smear, FBC, culture, (electrolytes).
5. **Glucose**: if <3.0 mmol/L (UCG; WHO <2.5, or <3.0 in SAM) or you cannot measure it: **10% dextrose 5 mL/kg IV** rapidly.
6. **If still fitting at 5 minutes**: first benzodiazepine dose (see Management). Start the clock for the 10-minute repeat.
7. **Temperature**: undress the child; paracetamol only once the fit has stopped and the child can swallow (or rectally).
8. **ETAT emergency signs**: shock (cold hands, CRT >3 s, weak fast pulse) → fluid bolus per WHO guidance (not in SAM without the SAM protocol); severe dehydration → rehydrate.

> **RED FLAG**
> Call the most senior person now if: the fit continues after **two benzodiazepine doses**, $SpO_2$ stays <90% or breathing is slow/irregular after diazepam, there are signs of **shock** or **raised ICP**, or the child is **<1 month old**.

## Step 1: Focused history

Take it from the eyewitness while the nurse sets up drugs. Each answer changes the plan:

- [ ] **Time the fit started and number of fits** → status? complex? severe malaria (≥2 in 24 h)?
- [ ] **Focal or generalised** → focal = complex → LP/imaging threshold drops.
- [ ] **Drugs given before arrival** (diazepam, antibiotics, antimalarials, herbs) → counts toward 2 benzodiazepine doses; partially treated meningitis.
- [ ] **Duration of fever and other symptoms** (vomiting, irritability, neck pain, rash, diarrhoea, cough) → meningitis, malaria, pneumonia, gastroenteritis.
- [ ] **Previous fits, development, head injury, sickle cell** → epilepsy, CP, stroke, trauma.
- [ ] **Immunisation (Hib, PCV)** → incomplete = higher meningitis risk.
- [ ] **HIV status, feeding, weight loss** → opportunistic CNS infections; SAM protocols.
- [ ] **Possible poisoning** → specific antidotes.

## Step 2: Focused examination

**General**: AVPU/Blantyre score, pallor, jaundice, rash, hydration, nutrition (MUAC, oedema).

**Vital signs**: temperature, RR, HR, CRT, $SpO_2$, BP. Repeat 15-minutely in the first hour.

**CNS**: pupils (size, symmetry, reaction), fontanelle, neck stiffness, Kernig/Brudzinski, tone and limb movements (symmetry; Todd's paresis), reflexes, posture, subtle seizure activity.

**Source**: ears (tympanic membranes), throat, chest (crackles, bronchial breathing), abdomen (splenomegaly in malaria; hepatomegaly), skin and joints.

| Document these positives | And these negatives |
|---|---|
| Fit duration, type, number and time of each drug | "No neck stiffness, fontanelle not bulging" |

| Document these positives | And these negatives |
|---|---|
| Temperature and source of fever | "Pupils equal and reactive, no focal deficit" |
| Glucose and malaria result | "No petechial rash" |
| Time back to full consciousness | "Blantyre 5/5 at 1 hour" |

## Step 3: Bedside tests

| Test | How | What it tells you |
|---|---|---|
| **Random blood glucose** | Glucometer, finger or heel prick | Hypoglycaemia (treat <3.0 mmol/L per UCG); also check after dextrose and 30 min later |
| **Malaria RDT** (and smear) | Finger prick; thick and thin film for parasite count | Malaria; parasitaemia + ≥2 fits/24 h or impaired consciousness = severe malaria |
| **Haemoglobin (HemoCue)** | Finger prick | Severe anaemia (Hb <5 g/dL) from malaria or sickle cell: transfuse if Hb ≤4 g/dL, or 4–6 g/dL with distress, shock or coma |
| **$SpO_2$** | Pulse oximeter | Hypoxaemia from the fit, aspiration or pneumonia |
| **Urine dipstick** | Clean-catch or bag (catheter if needed) | Nitrites/leucocytes in UTI (send culture) |
| **Bedside CSF look** | After LP | Cloudy CSF → treat as bacterial meningitis immediately |

## Step 4: Laboratory & imaging

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **Lumbar puncture: CSF cell count, Gram stain, protein, glucose (with paired blood glucose), culture; CrAg if HIV; Xpert if TB suspected** | Meningism, complex seizure, not normal within 1 h, <12 months (UCG: consider <2 years), incomplete Hib/PCV, pre-treated with antibiotics | Normal in febrile convulsion; bacterial meningitis: cloudy, high neutrophils, protein >0.4 g/L, glucose <1.5 mmol/L or CSF:blood ratio ≤0.4 | Ceftriaxone 100 mg/kg/day for 10–14 days. LP possible at HC IV (if trained) and hospitals; **culture only at RRH/NRH** (Mulago, Mengo labs) |
| **Blood smear with parasite count** | All febrile children | >250,000/µL or >10% = hyperparasitaemia | Severe malaria → IV artesunate. All levels |
| **FBC** | Complex seizure, ill child | Leucocytosis (bacterial), anaemia, thrombocytopenia (malaria, sepsis; low platelets → delay LP) | Antibiotics; transfusion. HC IV and above |
| **Blood culture** | Toxic child, suspected sepsis/meningitis | Organism | Tailor antibiotics. RRH/NRH; Mengo |
| **Electrolytes (Na, Ca, Mg), creatinine** | Diarrhoea/vomiting, prolonged fit, no fever source, abnormal recovery | Hypo/hypernatraemia, hypocalcaemia | Correct; reconsider the diagnosis (provoked seizure). RRH/NRH |
| **Urine microscopy and culture** | Infant with no clear source; dipstick positive | UTI | Antibiotics; ultrasound of kidneys if recurrent |
| **Chest X-ray** | Clinical pneumonia, hypoxaemia, possible aspiration | Consolidation | Antibiotic choice. Hospitals |
| **HIV test** (rapid test; DNA PCR if <18 months and exposed) | Unknown status | Positive | ART, co-trimoxazole, CrAg, think TB meningitis |

| Test | When | Expected / abnormal finding | How it changes management |
|---|---|---|---|
| **CT/MRI brain** | **Not** for simple febrile seizure. Consider for focal seizure with persistent deficit, signs of raised ICP, head injury, suspected abscess | Abscess, bleed, infarct, hydrocephalus | Neurosurgery. **CT at Mulago, Mengo and some RRHs/private centres; MRI rare and expensive** |
| **EEG** | **Not** for simple febrile seizure; consider for recurrent complex seizures, afebrile seizures, developmental concerns | Epileptiform discharges | Epilepsy diagnosis; specialist (Mulago, CoRSU, private) |

EXAM FAVOURITE

The AAP (2011) neurodiagnostic guideline for **simple** febrile seizures in children 6–60 months: **no routine** blood tests, EEG or neuroimaging; **LP if there are meningeal signs or symptoms or history suggesting meningitis**; LP is an **option** at 6–12 months if Hib/PCV immunisation is incomplete or unknown, and if the child has been pre-treated with antibiotics. In Uganda the threshold is lower because malaria and meningitis are common and follow-up is uncertain.

## Step 5: Putting it together

### Model one-line summary

> "An 18-month-old boy, fully immunised, HIV-negative, with a 6-hour history of fever and coryza, had a single 4-minute generalised tonic–clonic seizure at home and was fully alert 40 minutes later; temperature 39.4°C, no meningism, glucose 5.6 mmol/L, malaria RDT negative, red bulging right eardrum: first simple febrile convulsion secondary to acute otitis media."

### Problem list

1. Seizure with fever: simple febrile convulsion (diagnosis of exclusion: glucose normal, malaria negative, no meningism, rapid recovery).
2. Source: acute otitis media.
3. Parental anxiety: needs counselling and first-aid teaching.

EXAM FAVOURITE

**Worked example 1: interpreting the CSF.** A 10-month-old had a 20-minute focal fit and is still drowsy at 2 hours. CSF: cloudy; WBC 850/µL, 90% neutrophils; protein 1.8 g/L; glucose 0.9 mmol/L with blood glucose 6.0 (ratio 0.15); Gram-positive diplococci. **Interpretation:** pneumococcal meningitis. The "febrile convulsion" was an acute symptomatic seizure. Treat with ceftriaxone 100 mg/kg/day for 10–14 days, monitor for hearing loss, hydrocephalus and subdural effusions. Compare a **febrile convulsion**: clear CSF, WBC ≤5/µL, normal protein and glucose.

EXAM FAVOURITE

**Worked example 2: is it severe malaria?** A 3-year-old (13 kg) had 2 generalised fits in the last 12 hours; RDT positive, parasite count 80,000/µL; Blantyre 5/5 after 40 min; Hb 9.2 g/dL; glucose 4.8. **Interpretation:** by UCG 2023, **2 or more convulsions in 24 h with parasitaemia = severe malaria** even though the child is now alert. Give IV artesunate 3 mg/kg (39 mg) at 0, 12 and 24 h, then daily until able to take oral ACT; discharge on a full ACT course and **post-discharge malaria chemoprevention** (monthly DP for 3 months, UCG 2023).

# Differential diagnosis

| Condition | Distinguishing features | Key investigation |
|---|---|---|
| **Simple febrile convulsion** | 6 months–5 y, brief generalised fit, rapid full recovery, focus of fever outside CNS | Clinical; glucose, malaria test |
| **Cerebral malaria / severe malaria** | Malaria area, repeated or subtle fits, coma persisting >1 h after fit (Blantyre ≤2), anaemia, acidotic breathing, hypoglycaemia, retinopathy | RDT/smear, glucose, Hb; LP to exclude meningitis |
| **Bacterial meningitis** | Irritability, vomiting, bulging fontanelle, neck stiffness, petechiae, fits after day 1–2 of illness, focal fits, drowsiness | LP: cloudy CSF, neutrophils, low glucose; blood culture |
| **Viral encephalitis (HSV) / other encephalitis** | Behaviour change, focal fits, reduced consciousness for hours–days | CSF lymphocytes, PCR (rarely available), EEG, MRI |
| **TB meningitis** | Subacute (>5 days), weight loss, TB contact, HIV, cranial nerve palsies, hydrocephalus | CSF lymphocytes, high protein, low glucose; Xpert Ultra on CSF; CXR |
| **Hypoglycaemia** | Starved, vomiting, malaria, sepsis, SAM, quinine | Glucose <3.0 mmol/L |
| **Electrolyte disturbance** | Severe diarrhoea/dehydration (hypo/hypernatraemia), rickets (hypocalcaemia) | Na, Ca, Mg |
| **Epilepsy triggered by fever** | Previous afebrile seizures, abnormal development | History, EEG |
| **Shigella dysentery / gastroenteritis seizures** | Bloody diarrhoea, high fever | Stool, electrolytes |
| **Sickle cell stroke** | Known HbSS, hemiparesis, may be afebrile | Sickle test/Hb electrophoresis, CT |
| **Head injury / non-accidental injury** | History, bruising, retinal haemorrhages | CT head |
| **Poisoning** (organophosphates, paraffin, isoniazid, traditional herbs) | Exposure, pinpoint pupils, smell | History; specific antidote |
| **Rigors / febrile delirium / breath-holding / reflex anoxic seizure** | Conscious throughout, brief, triggered | Clinical |

# Investigations

| Test | What you are looking for | Why |
|---|---|---|
| Glucose | Hypoglycaemia | Treatable cause of fits and coma |

| Test | What you are looking for | Why |
|---|---|---|
| Malaria RDT/smear | Parasitaemia, density | Commonest cause of febrile fits in Uganda; severe malaria needs IV artesunate |
| LP (when indicated) | Meningitis/encephalitis | Missing meningitis is the most dangerous error |
| FBC / Hb | Anaemia, infection, platelets | Transfusion; LP safety |
| Urine dipstick/culture | UTI | Common occult source in infants |
| Electrolytes | Na, Ca | Provoked seizures |
| HIV test | Infection status | CNS opportunistic infection, ART |
| EEG / imaging | Only for complex/atypical cases | Not needed for simple febrile convulsions |

# Management

## A. Stopping the fit: the status epilepticus algorithm (child >1 month)

```
0 min     Fit starts. Time it. Recovery position, airway, O2, glucose check.
          Most fits stop by themselves within 5 minutes.
5 min     Still fitting -> 1st benzodiazepine:
            no IV: diazepam 0.5 mg/kg PR (max 10 mg)
                   or midazolam 0.3 mg/kg buccal/intranasal (max 10 mg)
            IV/IO: diazepam 0.2-0.3 mg/kg slow IV (max 10 mg)
                   (or midazolam 0.15 mg/kg IV, 0.2 mg/kg IM; max 10 mg)
          Glucose <3.0 or unknown -> 10% dextrose 5 mL/kg IV
15 min    Still fitting 10 min after dose 1 -> 2nd benzodiazepine (same dose)
          (if a dose was given at home/HC, this is already the 2nd: skip to 2nd line)
          Call senior. Prepare 2nd-line drug.
25 min    Still fitting = established status ->
            phenobarbital 15-20 mg/kg IV over 15-20 min (or IM 15 mg/kg)
            OR phenytoin 15-20 mg/kg IV in 0.9% saline over 20-60 min
            OR levetiracetam 40 mg/kg IV over 5 min (where available)
45 min    Still fitting = refractory -> give the other 2nd-line drug
          (e.g. phenytoin after phenobarbital; or extra phenobarbital 10 mg/kg)
          Anaesthetist / ICU: intubation, midazolam or thiopental infusion
```

DRUG DOSING

**Count every benzodiazepine dose**, including the one given at the drug shop or HC III. **No more than two doses** before a second-line drug: a third dose adds respiratory depression but little benefit. Have a bag-valve-mask ready whenever you give a benzodiazepine, especially after phenobarbital.

## B. Anticonvulsant drug table

| Drug | Dose | Route | Frequency / rate | Max dose | Notes |
|---|---|---|---|---|---|
| **Diazepam** | **0.5 mg/kg** | Rectal (10 mg/2 mL injection drawn into a 1-mL syringe, needle removed, inserted 4–5 cm) | Once; repeat once after 10 min if needed | **10 mg/dose**; 2 doses total | UCG 2023 and WHO Pocket Book (0.1 mL/kg). Hold the buttocks together for a few minutes |

| Drug | Dose | Route | Frequency / rate | Max dose | Notes |
|---|---|---|---|---|---|
| **Diazepam** | **0.2–0.3 mg/kg** (WHO: 0.25 mg/kg = 0.05 mL/kg) | IV, slowly over 1–2 min | Once; repeat once after 10 min | **10 mg/dose** | Apnoea risk; do not give IM (erratic absorption) |
| **Midazolam** | **0.3 mg/kg** (range 0.2–0.5) | Buccal (between gum and cheek) or intranasal | Once; repeat once after 10 min | **10 mg/dose** | NICE NG217 (2022) first-line in the community, as effective as rectal diazepam and more acceptable. Injection solution can be used buccally |
| **Midazolam** | **0.15 mg/kg** IV; **0.2 mg/kg** IM | IV / IM | Once; repeat once | **10 mg/dose** | IM useful if no IV and rectal not possible |
| **Phenobarbital** | **15–20 mg/kg** (WHO 15 mg/kg; UCG 10–15 mg/kg) | IV, diluted, over 15–20 min (≤1–2 mg/kg/min); or IM | Once; extra 10 mg/kg after 30 min if still fitting | **1 g** (single loading dose) | Widely available in Uganda; respiratory depression and hypotension, especially after benzodiazepines. Maintenance (if needed) 3–5 mg/kg/day |
| **Phenytoin** | **15–20 mg/kg** (UCG/WHO 15–18 mg/kg) | IV **in 0.9% saline only** (precipitates in dextrose), through a good separate line | Over **20–60 min**; rate **≤1 mg/kg/min** (max 50 mg/min) | Local protocols cap at 1–1.5 g (verify) | Cardiac monitor: bradycardia, arrhythmia, hypotension; extravasation causes necrosis ("purple glove"). Not IM. UCG: a further 10 mg/kg over 30 min if still fitting |
| **Levetiracetam** | **40 mg/kg** | IV over 5 min | Once | **3 g** | NICE NG217 (2022) suggests it as the preferred second-line option where available (fewer adverse effects, quick to give). Limited IV availability in Uganda |
| **Paraldehyde** (legacy) | 0.4 mL/kg | Rectal, mixed with equal volume of oil/saline | Once | 10 mL | Older WHO option; rarely stocked now |
| **Phenobarbital (neonate)** | **20 mg/kg** | IV (or IM) over 15–20 min | Extra 10 mg/kg after 30 min, up to **40 mg/kg** total | 40 mg/kg | **First line in neonates** (WHO); maintenance 2.5–5 mg/kg/day |
| **10% dextrose** | **5 mL/kg** | IV rapid bolus | Recheck glucose after 30 min | | Then maintenance fluid containing dextrose; feed as soon as safe |

UGANDA CONTEXT

**UCG 2023 (section 9.1.1, status epilepticus)** gives diazepam then **phenobarbital 10–15 mg/kg slowly IV** or **phenytoin 15–18 mg/kg over 1 hour**, then a further phenytoin 10 mg/kg. The child doses printed in that section ("0.05 mg/kg rectally, 0.02 mg/kg IV") appear to be misprints: the febrile convulsion section of the same guideline and the WHO Pocket Book both give **0.5 mg/kg rectally** (max 10 mg). Use 0.5 mg/kg PR and 0.2–0.3 mg/kg IV. On most Ugandan wards the sequence actually used is **diazepam → diazepam → phenobarbital → phenytoin**; levetiracetam IV and buccal midazolam are mainly available in private and some referral hospitals.

## C. Treat the cause and the fever

1. **Glucose**: correct hypoglycaemia (above); then IV fluids with dextrose at maintenance (Holliday–Segar 100 mL/kg/day for the first 10 kg, 50 mL/kg/day for the next 10 kg, 20 mL/kg/day thereafter) **restricted to about two-thirds** if meningitis or raised ICP is suspected; NG feeds as soon as safe.
2. **Malaria positive**:
   - Severe (≥2 fits in 24 h, impaired consciousness, prostration, etc.): **IV artesunate 3 mg/kg if <20 kg; 2.4 mg/kg if ≥20 kg** at 0, 12 and 24 h, then once daily until oral is tolerated, then a full course of oral ACT (UCG 2023; WHO malaria guidelines, latest consolidated edition). At a HC without IV facilities: **pre-referral rectal artesunate** (UCG: 100 mg for 5–<14 kg, 200 mg for 14–19 kg, children ≤6 years) and refer.
   - **Post-discharge malaria chemoprevention** (UCG 2023): monthly dihydroartemisinin–piperaquine for 3 months after severe malaria.
   - Uncomplicated (single brief fit, fully recovered): oral artemether–lumefantrine by weight band, observe for 24 h.
3. **Suspected bacterial meningitis** (or LP contraindicated/delayed): **ceftriaxone 100 mg/kg/day IV** (once daily or divided 12-hourly; max 4 g/day) for 10–14 days (UCG 2023). Do not delay antibiotics for an LP. Chloramphenicol 25 mg/kg 6-hourly if ceftriaxone unavailable. Under 2 months: ampicillin + gentamicin or ceftriaxone/cefotaxime per neonatal protocol.
4. **Other sources**: amoxicillin for otitis media or pneumonia, treat UTI, ORS/zinc for diarrhoea.
5. **Fever (for comfort, not to prevent fits)**:

| Drug | Dose | Route | Frequency | Max | Notes |
|---|---|---|---|---|---|
| **Paracetamol** | **15 mg/kg** | Oral (rectal if vomiting) | Every 6 h until fever settles (UCG) | **60 mg/kg/day**; 1 g/dose | Only once the child is awake and can swallow |
| **Ibuprofen** | 5–10 mg/kg | Oral | Every 6–8 h | 30–40 mg/kg/day | Avoid if dehydrated, in shock, <3 months; not with suspected dengue |

UGANDA CONTEXT

**UCG 2023 still recommends tepid sponging** for febrile convulsions and educating carers on "fever control". **NICE NG143** (fever in under 5s) states that **tepid sponging is not recommended** and **antipyretics do not prevent febrile convulsions**. A practical compromise on the ward: undress the child, give paracetamol for comfort, avoid cold water or alcohol sponging (which causes shivering and distress), and **tell parents clearly that a fit is not their fault** for not bringing the temperature down.

## D. Disposition

| Situation | Plan |
|---|---|
| First simple febrile convulsion, >12 months, alert, source found, parents able to return | Observe 4–6 hours (often overnight in Uganda because of distance), treat source, counsel, discharge |
| <12 months, complex seizure, not fully recovered, LP abnormal or not done, malaria-positive with ≥2 fits, social concerns | **Admit**; neuro-observations (Blantyre/GCS, pupils, glucose) 1–2-hourly for the first 6 hours |
| Status epilepticus, raised ICP, shock, respiratory failure | HDU/ICU (Mulago, Mengo HDU, RRHs with ICU) |
| HC II/III | Rectal diazepam, dextrose, pre-referral rectal artesunate and/or IM ceftriaxone (or benzylpenicillin + gentamicin), then refer with a note and an escort |

### E. Counselling the family

Parents often think the child is dying. Counsel in plain language (Luganda/English as appropriate), with both parents if possible:

1. **What happened**: "A fit triggered by a high temperature in a young child's brain. It is common (1 in 30 children) and does not damage the brain."
2. **Prognosis**: about **1 in 3** will have another, usually within the next year; most children **outgrow them by 5–6 years**. Epilepsy develops in only about **1–2 in 100** (only slightly more than in all children).
3. **First aid at home**:
   - Stay calm, **note the time**; put the child on their side on a soft, safe surface away from the fire.
   - **Do not put anything in the mouth** (no spoon, finger, herbs, sugar or water), **do not restrain**, do not pour water on the child, **no traditional burns or incisions**.
   - If it lasts **more than 5 minutes**, or the child turns blue, or has repeated fits, or does not wake up properly: go to the nearest health facility immediately (rectal diazepam given by a trained person, or a home rescue buccal midazolam plan for selected families with recurrent prolonged fits).
4. **Fever**: give paracetamol for comfort, plenty of fluids and breastfeeds, light clothing. **This will not necessarily prevent another fit**, so a fit is not their failure.
5. **When to come back**: drowsiness, persistent vomiting, stiff neck, rash, breathing difficulty, fits >5 minutes or repeated fits.
6. **Malaria prevention**: sleep under an insecticide-treated net; test and treat every fever early.
7. **Immunisation**: **continue** all UNEPI vaccines (including measles-rubella); febrile convulsions are not a contraindication.
8. **No daily anticonvulsants** are needed after simple febrile convulsions; intermittent diazepam during fevers is not routinely recommended (side effects outweigh benefits).
9. **Stigma**: febrile fits are **not epilepsy, not contagious, and not caused by witchcraft**; the child can go to day care and school normally.

> NOTE
> **Long-term anticonvulsants?** Not for simple febrile seizures (AAP 2008 guidance on long-term management; NICE NG217 2022). Consider specialist review for frequent or prolonged febrile seizures, focal features, developmental concerns or a strong family history of epilepsy (possible GEFS+ or Dravet syndrome; avoid sodium-channel blockers such as carbamazepine in suspected Dravet).

## Complications

| Of the seizure | Of treatment | Long-term |
|---|---|---|
| Aspiration pneumonia, hypoxia | Respiratory depression (benzodiazepines + phenobarbital) | Recurrence (~30%) |
| Injury, burns (fire), tongue bite | Hypotension, arrhythmia (phenytoin) | Epilepsy (1–2% after simple; higher after complex) |
| Hypoglycaemia, hyperthermia, acidosis | Extravasation injury (phenytoin, dextrose) | Mesial temporal sclerosis after febrile status (rare) |
| Status epilepticus → hippocampal injury | Sedation masking deterioration | Parental anxiety; stigma |

# Ward-round & exam pearls

### Model presentation

> "Baby K is an 18-month-old boy from Mukono, weighing 10.5 kg (WAZ −0.4, length 81 cm, HAZ −0.4, WHZ −0.2, MUAC 14.5 cm), fully immunised per the UNEPI card including three doses of DPT-HepB-Hib and PCV, HIV-unexposed, breastfeeding with a family diet. He presented 2 hours ago with a 6-hour history of fever and a runny nose, followed by one generalised tonic–clonic seizure lasting about 4 minutes at home, which stopped without treatment. He was sleepy for 30 minutes and is now playing. There is no previous history of seizures; development is normal (walks alone, says 6 words); an older sister had febrile fits. On examination, temperature 39.4°C, RR 32, HR 140, CRT <2 s, $SpO_2$ 98%, Blantyre 5/5, no meningism, fontanelle closed, no rash, red bulging right tympanic membrane. Glucose 5.6 mmol/L, malaria RDT negative. **Impression: first simple febrile convulsion secondary to right acute otitis media.** Plan: oral amoxicillin, paracetamol for comfort, observe overnight, counsel parents on first aid and recurrence, review in clinic if any further fits."

### Viva questions (with answers)

1. **Define a febrile convulsion.** Seizure with fever ≥38°C, age about 6 months to 5 years, no CNS infection, metabolic cause or previous afebrile seizure.
2. **Simple vs complex?** Simple: generalised, <15 min, once in 24 h, full recovery within 1 h. Complex: focal, ≥15 min, recurrent in 24 h, or slow recovery.
3. **When do you start anticonvulsants?** If the fit lasts 5 minutes or longer (ILAE 2015 t1), or repeated fits without recovery.
4. **Rectal diazepam dose?** 0.5 mg/kg (0.1 mL/kg of 10 mg/2 mL), max 10 mg; repeat once after 10 min.
5. **What after two doses?** Phenobarbital 15–20 mg/kg IV over 15–20 min (or IM) or phenytoin 15–20 mg/kg IV over 20–60 min in saline; levetiracetam 40 mg/kg if available.
6. **Indications for LP?** Meningism, complex seizure, incomplete recovery, <12 months (lower threshold <2 years in Uganda), incomplete Hib/PCV, prior antibiotics; not if raised ICP, shock, coagulopathy or local skin infection.
7. **Risk factors for recurrence?** Age <18 months, family history of febrile seizures, low peak temperature, short duration of fever before the fit.
8. **Do antipyretics prevent recurrence?** No.

### OSCE tips

- In a **counselling station**: introduce yourself, explore what the parent saw and fears ("Is my child going to die? Is it epilepsy? Was it witchcraft?"), explain in plain words, cover recurrence, first aid (with a demonstration of the recovery position), danger signs, no need for daily medicine, continue immunisation, check understanding, and give a written leaflet if available.
- In a **management station**: say "I would time the seizure, call for help, ABC, check glucose" before naming any drug. Calculate the dose aloud (e.g. 12 kg × 0.5 mg/kg = 6 mg = 1.2 mL of 10 mg/2 mL).

# Mnemonics & memory aids

- **"5-10-2"**: treat at **5** minutes, repeat after **10** minutes, maximum **2** benzodiazepine doses.
- **"DEFG": Don't Ever Forget Glucose** in a fitting or unconscious child.
- **Simple = "1-15-1"**: **1** fit, **<15** minutes, recovered within **1** hour (and generalised).

- **Recurrence risk factors "YFLS"**: **Y**oung (<18 months), **F**amily history, **L**ow temperature at the fit, **S**hort fever duration.
- **Before you say "febrile fit", exclude "MMGHE"**: **M**alaria (cerebral), **M**eningitis, **G**lucose, **H**ead injury/poisoning, **E**pilepsy/electrolytes.

# High-yield summary

### Numbers & doses to know

| Item | Value |
|---|---|
| Fever threshold | **≥38°C** |
| Age range | **6 months–5 years** (UCG 2023: 3 months–6 years); peak 12–18 months |
| Prevalence | **2–5%** of children |
| Simple febrile convulsion | Generalised, **<15 min**, once in **24 h**, recovered within **1 h** |
| Febrile status | **≥30 min** |
| Start anticonvulsant | Fit lasting **≥5 min** |
| Diazepam rectal | **0.5 mg/kg** (0.1 mL/kg of 10 mg/2 mL), **max 10 mg**, repeat once after 10 min |
| Diazepam IV | **0.2–0.3 mg/kg** slowly, max 10 mg |
| Midazolam buccal/intranasal | **0.3 mg/kg** (0.2–0.5), max 10 mg; IV 0.15 mg/kg, IM 0.2 mg/kg |
| Phenobarbital | **15–20 mg/kg** IV over 15–20 min or IM (max 1 g); neonates **20 mg/kg**, then 10 mg/kg, max 40 mg/kg |
| Phenytoin | **15–20 mg/kg** IV in saline, **≤1 mg/kg/min** (UCG/WHO over 60 min) |
| Levetiracetam | **40 mg/kg** IV over 5 min, max 3 g |
| Hypoglycaemia | <3.0 mmol/L (UCG); WHO <2.5 (<3.0 in SAM) → **10% dextrose 5 mL/kg** |
| Paracetamol | **15 mg/kg 6-hourly**, max 60 mg/kg/day |
| IV artesunate | **3 mg/kg** (<20 kg) or 2.4 mg/kg (≥20 kg) at 0, 12, 24 h then daily |
| Severe malaria (convulsions) | **≥2 convulsions in 24 h** with parasitaemia (UCG 2023) |
| Ceftriaxone (meningitis) | **100 mg/kg/day** (max 4 g) for 10–14 days |
| Recurrence | ~**30%**; ~50% if first fit <12 months; >70% with all 4 risk factors; <20% with none |
| Epilepsy after simple FS | ~**1–2%** (general population ~1%) |

### Classic exam traps

- **"Give paracetamol to prevent another fit."** Wrong: antipyretics do not prevent recurrence; give them for comfort.
- **"Malaria RDT positive, so no LP."** Wrong: parasitaemia is common; meningism or a low coma score still needs LP (or treatment for meningitis).
- **"A third dose of diazepam."** Wrong: after two doses move to phenobarbital/phenytoin/levetiracetam.
- **"Diazepam IM."** Wrong: erratic absorption; use rectal, buccal, IV, or IM midazolam.
- **"Phenytoin in dextrose."** Wrong: it precipitates; dilute only in 0.9% saline.
- **"Start daily phenobarbital after a simple febrile fit."** Wrong: no long-term anticonvulsant.

- **"A 4-month-old with fever and a fit has a febrile convulsion."** Wrong: treat as sepsis/meningitis; the definition starts at 6 months (UCG 3 months).
- **"CT and EEG for every first febrile fit."** Wrong: not for simple febrile seizures.

### Questions seniors ask

**The child is still fitting on arrival and the mother says she got "an injection in the bottom" at the clinic an hour ago. What do you do?** Assume a benzodiazepine (or quinine/IM antibiotic) may have been given; check glucose, give one dose of benzodiazepine (that would be the second), and prepare phenobarbital if the fit continues 10 minutes later.

**When would you NOT do an LP in a child you suspect has meningitis?** Signs of raised ICP (unequal pupils, posturing, focal deficit, irregular breathing, deteriorating coma score), shock or cardiorespiratory instability, ongoing seizures, low platelets or bleeding, infection at the LP site. Give ceftriaxone now and do the LP when safe.

**Why does a seizure need treatment at 5 minutes when we call status 30 minutes?** Seizures lasting more than 5 minutes are unlikely to stop on their own, and benzodiazepines become less effective as GABA-A receptors are internalised; damage becomes likely by 30 minutes (ILAE 2015 t1 and t2).

**How do you tell a post-ictal child from cerebral malaria coma?** Time course: the post-ictal child steadily improves and is fully alert within an hour. Persistent Blantyre ≤2 beyond 1 hour, retinopathy, hypoglycaemia, severe anaemia or acidosis point to cerebral malaria (and you must exclude meningitis).

**Will my child get epilepsy?** After a simple febrile seizure the risk is about 1–2 in 100, close to the risk for any child. It is higher after complex seizures, with a family history of epilepsy or with developmental problems.

**What is the recurrence risk and what raises it?** About 1 in 3 overall. Higher with first fit before 18 months, a family history of febrile seizures, a fit at a relatively low temperature, and a short time between fever onset and the fit.

**Should this child stop vaccines?** No. Vaccine-associated fever can trigger a febrile seizure, but the vaccines prevent the very infections (Hib, pneumococcus, measles) that cause meningitis and seizures.

SELF-TEST

# Flashcards

27 cards. Cover the right-hand column and test yourself.

| # | Question | Answer |
|---|---|---|
| 1 | Define a febrile convulsion. | A seizure with fever **≥38°C** in a child aged about **6 months to 5 years** (UCG 2023: 3 months to 6 years) **without** CNS infection, metabolic cause or a history of afebrile seizures. A **diagnosis of exclusion**. |
| 2 | Criteria for a SIMPLE febrile convulsion? | **Generalised**, lasts **<15 minutes**, **does not recur within 24 h** (same illness), and the child is **fully recovered within 1 hour** ('1-15-1'). |
| 3 | What makes a febrile convulsion COMPLEX? | Any of: **focal** onset or Todd's paresis, **≥15 minutes**, **recurrence within 24 h**/same illness, or prolonged altered consciousness (>1 h). |
| 4 | Febrile status epilepticus: definition? | A febrile seizure (or series without recovery) lasting **≥30 minutes**. |
| 5 | ILAE 2015 time points t1 and t2 for convulsive status epilepticus? | **t1 = 5 minutes** (start treatment: the seizure is unlikely to stop by itself). **t2 = 30 minutes** (risk of long-term neuronal injury). |
| 6 | Rectal diazepam dose for a fitting child? | **0.5 mg/kg** (= **0.1 mL/kg** of 10 mg/2 mL), **max 10 mg**; repeat once after 10 minutes (UCG 2023, WHO Pocket Book). |
| 7 | IV diazepam dose in a child? | **0.2–0.3 mg/kg** (WHO: 0.25 mg/kg = 0.05 mL/kg) slowly over 1–2 min, max 10 mg. Have a bag-valve-mask ready. |
| 8 | Buccal midazolam dose for a prolonged seizure? | **0.3 mg/kg** (range 0.2–0.5 mg/kg), **max 10 mg**; NICE NG217 (2022) first-line in the community. IV 0.15 mg/kg; IM 0.2 mg/kg. |
| 9 | Maximum number of benzodiazepine doses before a second-line drug? | **Two** (including any dose given before arrival). Then phenobarbital, phenytoin or levetiracetam. |
| 10 | Phenobarbital loading dose in status epilepticus (child >1 month)? | **15–20 mg/kg** IV over 15–20 min (or IM; WHO 15 mg/kg; UCG 10–15 mg/kg), max 1 g. Watch for apnoea and hypotension. |
| 11 | Phenytoin loading dose and how to give it? | **15–20 mg/kg IV in 0.9% saline only**, rate **≤1 mg/kg/min** (UCG/WHO: over 60 min), separate good line, cardiac monitoring. Not IM; not in dextrose. |
| 12 | Levetiracetam dose for established status epilepticus? | **40 mg/kg IV over 5 minutes, max 3 g** (NICE NG217 2022 suggests it as preferred second-line where available). |
| 13 | First-line anticonvulsant in a neonate? | **Phenobarbital 20 mg/kg** IV/IM; a further **10 mg/kg** after 30 min if needed, up to **40 mg/kg**; maintenance 2.5–5 mg/kg/day (WHO). |
| 14 | Treatment of hypoglycaemia in a fitting child? | **10% dextrose 5 mL/kg IV** rapidly, recheck glucose after 30 min, then dextrose-containing maintenance fluid and feeds. Threshold <3.0 mmol/L (UCG); WHO <2.5 (<3.0 in SAM). |
| 15 | UCG 2023 malaria criterion involving convulsions? | **2 or more convulsions in 24 hours** with parasitaemia = **severe malaria** → IV artesunate. |
| 16 | IV artesunate dose in a child? | **3 mg/kg if <20 kg; 2.4 mg/kg if ≥20 kg** at 0, 12 and 24 h, then once daily until oral is tolerated, then full oral ACT; post-discharge chemoprevention (monthly DP for 3 months, UCG 2023). |
| 17 | Four risk factors for recurrence of febrile seizures? | **Age <18 months**, **family history** of febrile seizures, **low peak temperature** at the fit, **short fever duration** before the fit. All 4 → >70%; none → <20%. |
| 18 | Overall recurrence risk after a first febrile convulsion? | About **30%** (about 50% if the first was before 12 months). |

| # | Question | Answer |
|---|---|---|
| 19 | Risk of epilepsy after simple febrile seizures? | About **1–2%** vs about 1% in all children. Higher after complex seizures, with a family history of epilepsy or neurodevelopmental problems. |
| 20 | Do antipyretics prevent febrile convulsions? | **No** (NICE NG143, RCH). Give paracetamol 15 mg/kg 6-hourly for comfort only; NICE also advises against tepid sponging. |
| 21 | AAP indications for LP after a simple febrile seizure? | **Meningeal signs or history suggesting meningitis**; an option at **6–12 months if Hib/PCV incomplete or unknown**; an option if **pre-treated with antibiotics**. Uganda: lower threshold (UCG: especially <2 years). |
| 22 | Contraindications to LP in a sick child (WHO)? | Signs of **raised ICP** (unequal pupils, rigid posture/posturing, limb paralysis, irregular breathing), **shock/instability**, **ongoing seizures**, bleeding tendency/low platelets, skin infection at the site. Give antibiotics; LP later. |
| 23 | Empirical antibiotic for suspected bacterial meningitis in a child (UCG 2023)? | **Ceftriaxone 100 mg/kg/day IV** (once daily or 12-hourly; max 4 g/day) for 10–14 days; chloramphenicol 25 mg/kg 6-hourly if unavailable. |
| 24 | Why must phenytoin not be diluted in dextrose? | It **precipitates** in glucose solutions. Use 0.9% saline and flush the line before and after. |
| 25 | Three first-aid 'don'ts' for parents during a febrile fit? | **Don't put anything in the mouth**, **don't restrain** the child, **don't pour water or apply traditional remedies/burns**. Do: time it, lie on side, seek care if >5 min. |
| 26 | Is a febrile convulsion a reason to stop immunisations? | **No.** Continue all UNEPI vaccines; they prevent the infections (Hib, pneumococcus, measles) that cause meningitis and seizures. |
| 27 | Are EEG and CT indicated after a first simple febrile seizure? | **No.** Reserve for complex/atypical seizures, persistent focal deficit, raised ICP or developmental concerns. |

SELF-TEST

# Single best answer questions

15 questions. The answers and explanations follow at the end.

**1.** A 2-year-old girl (12 kg) is carried into Mengo casualty fitting. Her mother says the generalised jerking started 7 minutes ago at home; nothing was given. There is no IV access yet. Airway is positioned and oxygen is on. Glucose is 4.8 mmol/L. What is the MOST appropriate next step?

- A Wait until the seizure has lasted 15 minutes before giving any drug
- B Give diazepam 6 mg rectally
- C Give phenobarbital 240 mg IV over 15 minutes
- D Give diazepam 6 mg intramuscularly
- E Give 10% dextrose 60 mL IV

**2.** A 15-month-old boy (10 kg) had a 3-minute generalised seizure at home 1 hour ago. He is now playful. Temperature 39.2°C, RR 34, no meningism, fontanelle soft, red bulging left eardrum. Glucose 5.2 mmol/L; malaria RDT negative. He is fully immunised. Which is the MOST appropriate plan?

- A Lumbar puncture, then IV ceftriaxone pending CSF results
- B CT scan of the head before discharge
- C Start oral phenobarbital 5 mg/kg/day for 6 months
- D Treat the otitis media, give paracetamol for comfort, observe and counsel the parents
- E Tepid sponge hourly and give paracetamol regularly to prevent another fit

**3.** A 3-year-old (14 kg) has been fitting for 25 minutes. She received rectal diazepam at a HC III 20 minutes ago and IV diazepam 4 mg on arrival 10 minutes ago. An IV line is running and glucose is 5.0 mmol/L. What is the MOST appropriate next drug?

- A Phenobarbital 280 mg IV over 15–20 minutes
- B A third dose of IV diazepam 4 mg
- C Phenytoin 280 mg IV in 5% dextrose over 5 minutes
- D Paracetamol 210 mg rectally
- E Midazolam 4 mg buccally

**4.** A 9-month-old boy (8 kg) had a 20-minute fit affecting the right arm and face. Two hours later he is irritable when handled and drowsy (Blantyre 3). Temperature 38.9°C; the fontanelle is full; pupils equal and reactive; no focal weakness; breathing regular; BP normal; platelets 210 × $10^9$/L. He received only 2 of 3 PCV doses. Malaria RDT is positive. What is the MOST appropriate next step? Advanced

- A Diagnose cerebral malaria and treat with IV artesunate alone
- B Discharge on oral artemether–lumefantrine after observation
- C CT scan of the head before any further treatment
- D Start daily oral carbamazepine
- E Perform a lumbar puncture and start IV ceftriaxone with IV artesunate

**5.** A 4-year-old (16 kg) with fever for 2 days had two generalised convulsions in the last 10 hours, each lasting about 3 minutes. He is now fully alert. Malaria smear: 60,000 parasites/µL; Hb 9.8 g/dL; glucose 5.5 mmol/L; no meningism. According to UCG 2023, what is the BEST management?

A Oral artemether–lumefantrine and discharge home
B Simple febrile convulsion: paracetamol and reassurance only
C IV artesunate 48 mg at 0, 12 and 24 h, then daily until oral ACT can be taken
D IV quinine as first line
E Blood transfusion and oral ACT

**6.** A 6-day-old baby (3 kg) born at home has fever and is having repeated jerking of the limbs with lip-smacking. Glucose is 4.2 mmol/L. An IV line is in. Which is the MOST appropriate anticonvulsant?

A Phenobarbital 60 mg IV over 15–20 minutes
B Diazepam 1.5 mg rectally
C Phenytoin 60 mg IV in dextrose
D Carbamazepine 15 mg orally
E Paraldehyde 1.2 mL rectally

**7.** You are about to do an LP on a 2-year-old with fever and a 25-minute fit 3 hours ago. He now has a Blantyre score of 2, the left pupil is larger than the right and sluggish, and he intermittently extends all four limbs. HR 70/min, BP 120/80 mmHg, breathing irregular. What is the MOST appropriate action? Advanced

A Proceed with the LP in the lateral position using a fine needle
B Perform the LP sitting up to reduce the risk of herniation
C Delay all antibiotics until a CT scan is done
D Do not perform the LP; give IV ceftriaxone (and antimalarial if indicated), nurse head-up 30° and seek senior/ICU help
E Give a rapid 20 mL/kg saline bolus to improve cerebral perfusion

**8.** Which child has the HIGHEST risk of a further febrile convulsion?

A A 4-year-old whose first fit occurred at 40.5°C after 2 days of fever, no family history
B A 13-month-old whose fit occurred at 38.4°C within 30 minutes of fever onset and whose brother had febrile fits
C A 3-year-old with a simple febrile convulsion caused by malaria
D A 2.5-year-old with a fit at 39.8°C after 24 hours of fever, no family history
E A 5-year-old with a fit after MR vaccination at 9 months who is now well

**9.** A mother of a 20-month-old who had a simple febrile convulsion yesterday asks what she should do if it happens again at home. Which advice is MOST appropriate?

A Put a spoon between his teeth so he does not bite his tongue
B Hold his arms and legs firmly until the jerking stops
C Pour cold water over him to bring the temperature down quickly
D Give him paracetamol by mouth during the fit
E Lay him on his side on a safe surface, note the time, and go to a health facility if it lasts more than 5 minutes

**10.** An 11-month-old girl (8.5 kg) had a 5-minute generalised fit and is fully alert after 40 minutes. Temperature 39°C; no focus on examination; RDT negative; glucose normal; no meningism. She has had only BCG and the birth OPV dose because the family moved districts. What is the MOST appropriate next step? Advanced

A Discharge with paracetamol; no further tests
B EEG within 48 hours
C Consider a lumbar puncture and examine the urine
D Start intermittent oral diazepam with every future fever
E Prescribe oral amoxicillin and discharge

**11.** A 3-year-old (15 kg) is in status epilepticus. After two doses of diazepam, phenytoin is chosen. Which prescription is correct?

A Phenytoin 300 mg IM into the thigh
B Phenytoin 300 mg in 0.9% saline IV over 20–60 minutes with cardiac monitoring
C Phenytoin 300 mg in 5% dextrose IV over 5 minutes
D Phenytoin 75 mg IV push
E Phenytoin 600 mg IV over 30 minutes

**12.** A 2-year-old (11 kg) with severe acute malnutrition (MUAC 10.8 cm, bilateral pitting oedema) is admitted after a generalised fit. Glucose is 2.7 mmol/L. What is the MOST appropriate immediate step? Advanced

A No treatment: glucose is above 2.5 mmol/L
B Give 50% dextrose 11 mL IV
C Give oral F-75 only after the next feed is due in 3 hours
D Start IV 0.9% saline 20 mL/kg
E Give 10% dextrose 55 mL IV, then feed and recheck glucose in 30 minutes

**13.** Which statement about febrile convulsions is TRUE?

A Children with simple febrile convulsions have a risk of epilepsy only slightly above that of the general population
B Regular paracetamol during fevers prevents recurrence
C Most simple febrile convulsions cause lasting learning difficulties
D A first simple febrile convulsion needs an EEG
E The peak age is 4–5 years

**14.** A 7-year-old girl (22 kg) with known sickle cell anaemia presents with fever 38.6°C and a 10-minute fit that started with jerking of the right arm. One hour later she cannot lift her right arm or leg and her speech is slurred. Malaria RDT negative; glucose normal. What is the MOST likely diagnosis? Advanced

A Complex febrile convulsion with Todd's paresis
B Simple febrile convulsion
C Cerebral malaria
D Acute ischaemic stroke complicating sickle cell disease
E Breath-holding attack

**15.** A HC III in Wakiso receives a 3-year-old (14 kg) who is drowsy after two fits today, has a positive malaria RDT and neck stiffness. There are no IV facilities. The child has stopped fitting. What is the BEST pre-referral treatment?

A Oral artemether–lumefantrine and paracetamol, then refer
B Rectal diazepam 7 mg and refer without antimalarial or antibiotic
C Rectal artesunate 200 mg plus IM ceftriaxone, then urgent referral
D Keep the child for 24 hours of observation before referring
E IM quinine and oral amoxicillin, then refer the next morning

## Answers & explanations

**1. B** · Give diazepam 6 mg rectally

B is right: a seizure lasting ≥5 minutes needs a benzodiazepine; with no IV access give **diazepam 0.5 mg/kg rectally** (12 kg × 0.5 = 6 mg, max 10 mg). A is wrong: treatment starts at 5 minutes (ILAE t1), not 15; waiting makes benzodiazepines less effective. C is wrong: phenobarbital is second-line after two benzodiazepine doses, and there is no IV line. D is wrong: IM diazepam is absorbed erratically. E is wrong: glucose is normal (4.8 mmol/L), so dextrose is not indicated.

**2. D** · Treat the otitis media, give paracetamol for comfort, observe and counsel the parents

D is right: this is a **simple febrile convulsion** (generalised, <15 min, single, full recovery) with a clear source (otitis media) in a fully immunised child; treat the source, give paracetamol for comfort, observe and counsel. A is wrong: no meningeal signs, full recovery, immunised and >12 months: LP is not indicated. B is wrong: imaging is not indicated for simple febrile seizures. C is wrong: long-term anticonvulsants are not recommended after simple febrile seizures. E is wrong: antipyretics and sponging do not prevent recurrence (NICE NG143).

**3. A** · Phenobarbital 280 mg IV over 15–20 minutes

A is right: she has had **two benzodiazepine doses** (one at the HC III counts), so she needs a second-line drug: **phenobarbital 20 mg/kg** (14 × 20 = 280 mg) IV over 15–20 minutes. B is wrong: a third benzodiazepine dose adds respiratory depression with little benefit. C is wrong: phenytoin precipitates in dextrose and must be given at ≤1 mg/kg/min, not over 5 minutes. D is wrong: paracetamol does not stop seizures. E is wrong: midazolam is another benzodiazepine; the maximum of two has been reached.

**4. E** · Perform a lumbar puncture and start IV ceftriaxone with IV artesunate

E is right: this is a **complex** seizure (focal, 20 min) in an infant with a full fontanelle, irritability, incomplete PCV and slow recovery: **meningitis must be excluded** even though the RDT is positive. There are no signs of raised ICP or shock, so do the LP and treat both (ceftriaxone 100 mg/kg/day + IV artesunate). A is wrong: parasitaemia is common and does not exclude meningitis. B is wrong: a drowsy infant after a complex seizure needs admission. C is wrong: CT would delay treatment and is not routinely needed without raised ICP signs. D is wrong: there is no indication for maintenance anticonvulsants.

**5. C** · IV artesunate 48 mg at 0, 12 and 24 h, then daily until oral ACT can be taken

C is right: UCG 2023 defines **2 or more convulsions in 24 h with parasitaemia** as severe malaria; IV artesunate **3 mg/kg** for a child <20 kg (16 × 3 = 48 mg) at 0, 12, 24 h then daily, followed by a full oral ACT course. A is wrong: severe malaria must not be treated with oral ACT alone. B is wrong: two fits in 24 h is by definition complex, and malaria explains them. D is wrong: quinine is only used when artesunate is unavailable. E is wrong: Hb 9.8 g/dL does not need transfusion (transfuse at Hb ≤4 g/dL, or 4–6 g/dL with respiratory distress, shock or impaired consciousness).

**6. A** · Phenobarbital 60 mg IV over 15–20 minutes

A is right: in neonates **phenobarbital 20 mg/kg** (3 × 20 = 60 mg) is first line (WHO Pocket Book), with a further 10 mg/kg after 30 min if needed (max 40 mg/kg). This baby also needs a full sepsis/meningitis workup and IV antibiotics. B is wrong: WHO advises phenobarbital instead of diazepam in young neonates. C is wrong: phenytoin is not first line and must never be diluted in dextrose. D is wrong: oral maintenance drugs do not stop an acute seizure. E is wrong: paraldehyde is a legacy drug and not first-line in neonates.

**7. D** · Do not perform the LP; give IV ceftriaxone (and antimalarial if indicated), nurse head-up 30° and seek senior/ICU help

D is right: unequal pupils, posturing, bradycardia with hypertension and irregular breathing are **signs of raised ICP**, a contraindication to LP (WHO). Treat empirically now and support ICP (head-up 30°, avoid hypotonic/excess fluid, airway). A and B are wrong: LP in raised ICP risks coning, whatever the position or needle. C is wrong: antibiotics must not be delayed for imaging. E is wrong: he is not in shock; a large bolus may worsen cerebral oedema.

**8. B** · A 13-month-old whose fit occurred at 38.4°C within 30 minutes of fever onset and whose brother had febrile fits

B is right: this child has **all four recurrence risk factors**: age <18 months, family history, low temperature at the fit, and short fever duration (risk >70%). A is wrong: older age, high temperature and long fever duration are all low-risk features. C is wrong: the cause of fever (malaria) is not a listed risk factor and age 3 is lower risk. D is wrong: no risk factors (risk <20%). E is wrong: a vaccine-associated fit years ago at a now-older age does not predict recurrence; the child is past the usual age range.

**9. E** · Lay him on his side on a safe surface, note the time, and go to a health facility if it lasts more than 5 minutes

E is right: the recovery position, timing the fit and seeking care if it lasts >5 minutes (or recurs, or he does not wake properly) is correct first aid. A is wrong: objects in the mouth cause injury and aspiration. B is wrong: restraint can injure the child and does not stop the fit. C is wrong: cold water causes distress and does not stop the seizure. D is wrong: nothing should be given by mouth during a fit (aspiration risk).

**10. C** · Consider a lumbar puncture and examine the urine

C is right: she is **6–12 months old and has had no Hib or PCV doses**, so LP is reasonable (AAP option; lower threshold in Uganda), and with no focus a **urine test** is needed to exclude UTI. Catch-up immunisation should follow. A is wrong: with no immunisation and no focus, she needs more assessment. B is wrong: EEG is not indicated after a simple febrile seizure. D is wrong: intermittent diazepam prophylaxis is not routinely recommended. E is wrong: blind antibiotics without a diagnosis may partially treat meningitis and mask it.

**11. B** · Phenytoin 300 mg in 0.9% saline IV over 20–60 minutes with cardiac monitoring

B is right: phenytoin **20 mg/kg** (15 × 20 = 300 mg) in **0.9% saline** at **≤1 mg/kg/min** (20–60 min; UCG/WHO over 60 min) with cardiac monitoring through a secure line. A is wrong: IM phenytoin is erratically absorbed and causes tissue necrosis. C is wrong: phenytoin precipitates in dextrose and 5 minutes is far too fast (arrhythmia, hypotension). D is wrong: 5 mg/kg is an inadequate loading dose and a push is dangerous. E is wrong: 40 mg/kg is a double dose.

**12. E** · Give 10% dextrose 55 mL IV, then feed and recheck glucose in 30 minutes

E is right: in severe malnutrition WHO treats hypoglycaemia at **<3.0 mmol/L** (UCG also uses <3.0): **10% dextrose 5 mL/kg** (11 × 5 = 55 mL) IV, then early feeds and recheck. A is wrong: the 2.5 mmol/L threshold does not apply in SAM. B is wrong: 50% dextrose is hypertonic and damages veins in children; use 10%. C is wrong: a symptomatic, fitting child needs IV glucose now, not a delayed feed. D is wrong: a saline bolus does not treat hypoglycaemia and risks heart failure in SAM.

**13. A** · Children with simple febrile convulsions have a risk of epilepsy only slightly above that of the general population

A is right: the epilepsy risk after simple febrile seizures is about 1–2% compared with about 1% in all children. B is wrong: antipyretics do not prevent recurrence. C is wrong: simple febrile seizures do not affect behaviour, school performance or cognition. D is wrong: EEG is not indicated. E is wrong: the peak age is 12–18 months.

**14. D** · Acute ischaemic stroke complicating sickle cell disease

D is right: a child with **HbSS** and a new **hemiparesis with dysphasia** has a stroke until proven otherwise; at 7 years she is also outside the febrile convulsion age range. She needs urgent imaging and exchange/simple transfusion per protocol. A is wrong: Todd's paresis is possible, but she is too old for febrile convulsions and the sickle cell history makes stroke the priority diagnosis. B is wrong: a focal 10-minute fit with a deficit is not simple, and the age is wrong. C is wrong: the RDT is negative and the picture is focal. E is wrong: breath-holding attacks are brief, triggered by crying, and do not cause hemiparesis.

**15. C** · Rectal artesunate 200 mg plus IM ceftriaxone, then urgent referral

C is right: UCG 2023 advises **pre-referral rectal artesunate** (14–19 kg: 2 × 100 mg suppositories = 200 mg) where IV treatment is not possible, plus a **pre-referral antibiotic** for suspected meningitis, then urgent referral. A is wrong: a drowsy child cannot safely take oral drugs and severe malaria needs parenteral treatment. B is wrong: the fits have stopped, so diazepam is not needed now, and omitting the antimalarial and antibiotic delays life-saving treatment. D is wrong: delaying referral is dangerous. E is wrong: oral amoxicillin does not treat meningitis, and referral must not wait until the morning.

SELF-TEST

# Clinical cases

Work through each step before reading the model answer.

## Case 1: The first fit: a frightened mother at Mengo

Nakato brings her 18-month-old son Brian to Mengo Hospital casualty at 9 pm. One hour ago he 'shook all over and his eyes rolled up' for about 4 minutes at home. He had a runny nose and fever since the afternoon. He is now sleepy but crying when examined.

**1. What are your first actions and what must you check before calling this a febrile convulsion?**

> ETAT ABCD: airway, breathing ($SpO_2$), circulation, coma score and convulsion status, dehydration. Vital signs. **Glucose** (DEFG). **Malaria RDT**. Assess consciousness (AVPU/Blantyre) and look for **meningism**, a bulging fontanelle and a **rash**. Febrile convulsion is a diagnosis of exclusion.

**2. What history do you take from Nakato?**

> Exact description (generalised vs focal), duration, number of fits, recovery; fever duration and source symptoms (cough, ear pulling, diarrhoea, urine); vomiting, irritability, neck pain, rash; drugs given (diazepam, antibiotics, herbs); previous fits; development; family history of febrile fits/epilepsy; **UNEPI card** (Hib, PCV); HIV status; feeding and recent weight; possible poisoning.

**3. Findings: weight 10.5 kg (WAZ −0.4), T 39.4°C, RR 32, HR 140, $SpO_2$ 98%, Blantyre 5/5 at 1 h, no meningism, red bulging right eardrum. Glucose 5.6 mmol/L, RDT negative. Fully immunised. Sister had febrile fits. What is your diagnosis?**

> **First simple febrile convulsion** (generalised, <15 min, single, full recovery within 1 h) secondary to **right acute otitis media**. No LP needed: >12 months, immunised, no meningism, rapid recovery.

**4. What treatment do you give?**

> **Amoxicillin** for otitis media (per UCG dosing by weight), **paracetamol 15 mg/kg** (about 160 mg) 6-hourly for comfort, fluids/breastfeeds, light clothing. No anticonvulsant maintenance. Observe overnight (distance, first fit).

**5. What is his risk of another febrile fit?**

> He has **two** risk factors (age <18 months and family history), so his risk is higher than the average of about 30%; perhaps around 1 in 2. Explain this honestly but reassure that further simple fits do not harm the brain.

**6. How do you counsel Nakato before discharge?**

> Explain what a febrile fit is, that it is common and benign, not epilepsy, not witchcraft. Recurrence about 1 in 3 (higher for Brian). Epilepsy risk only about 1–2%. First aid: side position, time it, nothing in the mouth, no restraint or water; go to a facility if >5 min, repeated, or not waking. Paracetamol is for comfort and will not necessarily prevent fits. Continue immunisations; use a mosquito net. Return if drowsy, vomiting, stiff neck or rash.

**Take-away:** A simple febrile convulsion is diagnosed only after glucose, malaria and meningitis are considered; then the main treatment is the source of fever plus good counselling.

## Case 2: Still fitting after two doses

Okello, a 3-year-old boy (14 kg) from Luwero, arrives at the paediatric emergency unit fitting. His father says the fit began 25 minutes ago; at the HC IV he was given rectal diazepam 15 minutes ago. He has had fever for 2 days.

**1. What do you do in the first 2 minutes?**

> Note the times, call for help. Recovery position, suction, **oxygen**, bag-valve-mask ready. Check **glucose** at once. IV/IO access; draw blood for RDT/smear, FBC, culture. Count benzodiazepine doses: **one already given**.

**2. Glucose is 2.1 mmol/L. What do you give?**

> **10% dextrose 5 mL/kg = 70 mL IV** rapidly; recheck in 30 minutes; then dextrose-containing maintenance fluid. Hypoglycaemia is common in malaria and fasting children and can itself cause fits.

**3. He is still fitting. Which anticonvulsant now, and at what dose?**

> This will be his **second** benzodiazepine: **IV diazepam 0.2–0.3 mg/kg = about 3.5 mg** (0.25 mg/kg) slowly (or buccal midazolam 0.3 mg/kg = 4.2 mg). Prepare phenobarbital now.

**4. Ten minutes later he is still fitting. What next?**

> Established status: **phenobarbital 20 mg/kg = 280 mg IV** diluted over 15–20 min (or IM if no line; UCG 10–15 mg/kg), monitoring breathing and BP. Alternative: phenytoin 20 mg/kg (280 mg) in saline over 20–60 min, or levetiracetam 40 mg/kg (560 mg) if available. **No third benzodiazepine.**

**5. The fit stops. RDT positive, parasites 300,000/µL, Hb 6.5 g/dL, Blantyre 2 at 1 hour. What are the diagnosis and treatment?**

> **Severe (cerebral) malaria** with hyperparasitaemia, hypoglycaemia and repeated fits. **IV artesunate 3 mg/kg = 42 mg** at 0, 12, 24 h then daily; maintain glucose; transfuse if Hb falls below 5 g/dL or with complications per protocol. Meningitis is not excluded: if no raised-ICP signs do the LP, otherwise start **ceftriaxone 100 mg/kg/day** empirically.

**6. What monitoring and follow-up are needed?**

> Neuro-observations (Blantyre, pupils) and glucose 1–2-hourly, vital signs, fluid balance, watch for subtle seizures. On recovery: oral ACT course, **post-discharge malaria chemoprevention** (monthly DP for 3 months), hearing/development check, net use, and review for neurological sequelae (cerebral malaria can cause epilepsy, CP-like deficits, behaviour problems).

**Take-away:** Count every benzodiazepine dose, never give more than two, and always treat glucose; in Uganda the 'febrile fit' that does not stop is often cerebral malaria.

## Case 3: The infant who stays drowsy

Amina, a 9-month-old girl (7.8 kg, WAZ –1.2), is brought to a regional referral hospital after a 20-minute fit involving the left side of her body. She has had fever and vomiting for 3 days and was given an antibiotic syrup from a drug shop yesterday. Two hours later she is still drowsy and irritable when handled.

**1. Why is this not a simple febrile convulsion?**

> It is **focal**, lasted **≥15 minutes**, and she has **not recovered within 1 hour**: a **complex** seizure. Add: 3 days of illness, vomiting, irritability, prior antibiotics. These are red flags for **bacterial meningitis**.

**2. What examination findings would you look for before an LP?**

> Fontanelle (bulging?), neck stiffness (unreliable at 9 months), petechial rash, pupils, posturing, focal weakness, breathing pattern, HR/BP (Cushing response), CRT and shock, coma score, and platelets/bleeding. Any **raised-ICP sign or shock = no LP now**.

**3. Fontanelle is bulging; pupils equal; no posturing; breathing regular; BP normal. RDT negative. What next?**

> **Lumbar puncture** (no contraindication), then **ceftriaxone 100 mg/kg/day IV** (780 mg/day) immediately afterwards without waiting for results. Send CSF for cells, Gram stain, protein, glucose (with blood glucose), culture; blood culture; FBC; glucose; HIV test.

**4. CSF: turbid, WBC 1,200/µL (85% neutrophils), protein 2.1 g/L, glucose 0.8 mmol/L (blood 5.4). Interpret this, given the prior antibiotic.**

> **Bacterial meningitis** (neutrophilic, high protein, CSF:blood glucose ratio 0.15). Prior antibiotics may make the Gram stain and culture negative but do not normalise the cells and glucose this quickly. Treat for **10–14 days**; adjust to culture.

**5. What supportive care does she need?**

> Treat seizures (benzodiazepine → phenobarbital), glucose monitoring, fluids at about two-thirds maintenance if no shock (watch sodium), NG feeds when safe, antipyretic for comfort, neuro-observations, head circumference daily. Dexamethasone is not routinely used in low-income settings (no benefit shown in African trials).

**6. What follow-up do you arrange?**

> **Hearing test** (sensorineural deafness after pneumococcal/Hib meningitis), developmental follow-up, head circumference (hydrocephalus, subdural effusion), seizure follow-up, and **catch-up immunisation** (PCV, DPT-HepB-Hib).

> **Take-away:** A complex seizure with slow recovery in an infant, especially after drug-shop antibiotics, is meningitis until the CSF proves otherwise.

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A study aid, not a protocol. Check doses against the Uganda Clinical Guidelines 2023, your unit's protocol and a senior before you treat a patient.